COMPUTER LABORATORY • ASSIGNMENT 13

TIME SERIES DATA ANALYSIS

Stock Market Data • Apple (AAPL) closing prices

DETAILED CODE EXPLANATION + VIVA QUESTION BANK

Beginner-friendly theory, code walkthrough, function reference and cross-questions.

What your notebook does

Loads historical Apple stock data; prepares dates; plots closing prices; calculates 20- and 50-observation moving averages; aggregates prices monthly; decomposes the series; checks price-volume correlation; and forecasts 30 future observations using Exponential Smoothing.

Scope: This guide follows the submitted notebook. It clearly separates completed code from extra theory and possible improvements. The notebook uses Exponential Smoothing, not ARIMA.

1. Assignment overview and core theory

A **time series** is a sequence of observations recorded in time order. Examples: daily stock prices, hourly temperatures, monthly sales. In this notebook, one observation is a trading session and the main variable is AAPL.Close.

Concept	Simple explanation	In this assignment
Observation	One recorded data point	Closing price for one trading date
Trend	Long-term direction	Overall upward/downward price movement
Seasonality	Pattern repeating at a fixed interval	A repeating pattern if one exists
Noise / residual	Unexplained irregular movement	Unexpected day-to-day price changes
Moving average	Average over a fixed recent window	MA20 and MA50
Forecast	Estimate of a future value	Next 30 observations

Four common components

• **Level:** typical value around which a series moves.
• **Trend:** long-term direction.
• **Seasonality:** repeated pattern at a fixed interval.
• **Residual/noise:** what remains after systematic components are removed.

Why time order matters

The past is used to predict the future. Sorting dates is essential, and a random train-test split is usually inappropriate because it can mix future observations into training. A chronological split better imitates real forecasting.

Viva-ready summary

“I analyse Apple's historical closing prices, smooth fluctuations with moving averages, inspect monthly and decomposed patterns, study correlation with trading volume, and forecast future values using Exponential Smoothing.”

2. Libraries, loading and inspection

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.holtwinters import ExponentialSmoothing
```

Import	Purpose	Use in notebook
pandas (pd)	Table and date operations	CSV, sorting, index, resampling, rolling
numpy (np)	Numerical arrays and maths	Imported; no direct np call appears
matplotlib.pyplot (plt)	Plotting	Charts and show()
seaborn (sns)	Statistical charts	Scatter plot
seasonal_decompose	Split series into components	Trend, seasonal, residual
ExponentialSmoothing	Forecasting model	Fit additive-trend model

Code cell — load data

```
stocks = pd.read_csv('Stock_Prices.csv')
stocks.head()
```

pd.read_csv() reads a CSV file and returns a pandas DataFrame. The path is relative, so the CSV must be in the current working directory or a full path must be supplied. **stocks** stores the table. **head()** displays the first five rows by default; head(10) displays ten. **tail()** shows the last rows.

DataFrame versus Series

A DataFrame is a labelled 2-D table with rows and columns. A Series is one labelled column, such as stocks['AAPL.Close']. The notebook's data includes Date, AAPL.Open, AAPL.High, AAPL.Low, AAPL.Close, AAPL.Volume and AAPL.Adjusted.

Cross-questions

Why use aliases? Short names such as pd and plt reduce typing; they are conventions.
What if the CSV path is wrong? Python raises a file-not-found error.
What does head() return? A preview of the first rows, not the entire dataset.

3. Date preparation and historical price plot

```
stocks['Date'] = pd.to_datetime(stocks['Date'])
stocks = stocks.sort_values('Date')
stocks = stocks.set_index('Date')
```

Statement	Meaning	Reason
stocks['Date']	Select Date column	Access the date values
pd.to_datetime()	Convert values to datetime	Enables calendar-aware operations
sort_values('Date')	Sort rows chronologically	Keeps time order correct
set_index('Date')	Make Date the index	Enables resampling and time plotting

After set_index, Date is no longer an ordinary column; it becomes the row index. Sorting avoids lines jumping backward in time and ensures rolling windows follow the correct sequence.

Plot closing price

```
stocks['AAPL.Close'].plot(figsize=(13, 4))
```

This selects the closing-price Series and plots it. figsize=(13,4) sets width and height in inches. Date appears on the x-axis because it is the index; closing price appears on the y-axis. The chart helps reveal trends, sudden movements and volatility.

Useful definitions

Closing price: last traded price for the session. **Volatility:** degree of fluctuation in price. A chart may suggest volatility, but this notebook does not calculate a formal volatility measure.

Possible examiner follow-up

Why convert date before setting index? A datetime index is required for reliable resampling and date-aware operations.
What does figsize mean? Figure dimensions in inches.
What is sort_values? A pandas method that sorts rows by selected column values.

4. Moving averages and monthly aggregation

```
stocks['MA20'] = stocks['AAPL.Close'].rolling(20).mean()
stocks['MA50'] = stocks['AAPL.Close'].rolling(50).mean()
stocks[['AAPL.Close', 'MA20', 'MA50']].plot(figsize=(13, 4))
```

rolling(20) creates a window of 20 observations. **mean()** calculates the average in that window. MA20 and MA50 are new columns added to the DataFrame. The first 19 MA20 values and first 49 MA50 values are missing because a full window is not yet available.

Question	Answer
What does MA20 mean?	Average of the current and preceding 19 observations.
Is 20 equal to 20 calendar days?	No. It means 20 rows/observations; stock sessions skip weekends and holidays.
Why is MA50 smoother?	It averages more observations and reacts more slowly.
What is lag?	A smoothed curve responds later than the original series.
Can rolling use median?	Yes, for example rolling(20).median().

Monthly aggregation

```
monthly = stocks['AAPL.Close'].resample('ME').mean()
monthly.plot(kind='bar', figsize=(12, 4))
plt.show()
```

resample('ME') groups datetime-indexed observations into month-end calendar buckets. **mean()** computes the average closing price for each month. It does *not* select the final closing price of each month. kind='bar' requests bars; plt.show() displays the figure.

Rolling vs resampling: rolling moves a fixed-size window row by row; resampling groups data into calendar periods and usually returns one summary per period.

5. Seasonal decomposition and correlation

```
res = seasonal_decompose(
    stocks['AAPL.Close'],
    model='additive',
    period=20
)
res.plot()
plt.show()
```

Decomposition separates a series into components. Additive form: **Observed = Trend + Seasonal + Residual**. Observed is original data; trend is long-term movement; seasonal is the repeating component; residual is what remains.

period=20 tells the function to use a 20-observation cycle. For daily market data this is roughly 20 trading observations, not 20 calendar days. This parameter is an assumption; it does not prove that a true 20-session seasonal pattern exists.

Additive vs multiplicative

Additive is suitable when seasonal variation is roughly constant in size. Multiplicative uses Trend × Seasonal × Residual and is suitable when seasonal variation changes in proportion to the level. Multiplicative decomposition generally requires positive values.

Correlation code

```
print(stocks[['AAPL.Close', 'AAPL.Volume']].corr())
sns.scatterplot(x=stocks['AAPL.Close'], y=stocks['AAPL.Volume'])
```

Double brackets select multiple columns as a DataFrame. corr() calculates pairwise correlations; by default pandas uses Pearson correlation. The output in your notebook gives price-volume correlation **0.023473**, very close to zero: almost no linear association in these observations.

Coefficient	Interpretation
+1	Perfect positive linear relationship
Near 0	Little or no linear association
-1	Perfect negative linear relationship

Scatterplot shows each observation as a point and helps reveal clusters, outliers or curved patterns. Correlation does not prove causation and Pearson correlation may miss nonlinear relationships.

6. Forecasting with Exponential Smoothing

```
train = stocks['AAPL.Close'][:-30]
test = stocks['AAPL.Close'][-30:]

model = ExponentialSmoothing(train, trend='add').fit()
forecast = model.forecast(30)

plt.plot(train, label='Training Data')
plt.plot(test, label='Actual Price')
plt.plot(forecast, label='Forecast')
plt.show()
```

[:-30] selects all values except the last 30. **[-30:]** selects the last 30 values. Earlier data trains the model; the final 30 are held out for testing. Keeping the split chronological imitates real forecasting and avoids future information leaking into training.

ExponentialSmoothing(train, trend='add') creates a smoothing model with an additive trend. **.fit()** estimates model parameters from training observations. **.forecast(30)** generates the next 30 predictions. Exponential smoothing gives more weight to recent observations; the trend model estimates level and trend.

The plot shows training data, actual held-out prices and forecast values. The labels are attached to lines, but the notebook does not call plt.legend(); adding it would display the labels.

Critical limitation

The notebook does not calculate MAE, RMSE or another numerical forecast metric. A visual comparison alone is not enough to claim that the model is accurate. MAE is mean absolute error; RMSE is the square root of mean squared error. Lower values generally indicate smaller errors on the same scale.

The notebook output also warns that the date index has no associated frequency. Stock data skips weekends and holidays, so statsmodels may not infer a regular frequency and may use integer forecast indexes. Forecast values are generated, but date labels need careful handling.

7. Function reference and improvements

Function	What it does	Common cross-question
pd.read_csv()	Read CSV into DataFrame	Wrong path causes file-not-found error.
head() / tail()	Preview first / last rows	head() defaults to 5.
pd.to_datetime()	Convert to datetime	Invalid values can be coerced to NaT.
sort_values()	Sort rows by column	ascending=False reverses order.
set_index()	Set row index	reset_index() restores index as a column.
plot()	Create chart	figsize controls dimensions.
rolling()	Create moving window	20 means 20 observations.
mean()	Arithmetic average	Outliers can affect mean.
resample()	Group by time period	Needs datetime-like index.
show()	Display plot	Especially useful in scripts.
seasonal_decompose()	Split series components	Needs a sensible period and enough data.
corr()	Correlation matrix	Default Pearson correlation.
sns.scatterplot()	Scatter chart	Shows relationship and outliers.
ExponentialSmoothing()	Create forecasting model	trend='add' selects additive trend.
fit() / forecast()	Estimate / predict	forecast(30) returns 30 steps.

Improvements you can suggest in viva

1. **Handle frequency warning:** define an appropriate trading-session index and handle holidays/missing sessions carefully; do not blindly treat stock data as continuous calendar days.
2. **Evaluate:** calculate MAE and RMSE using test and forecast.
3. **Add legend:** call plt.legend() before plt.show().
4. **Check stationarity:** inspect whether mean/variance are stable; transformations may be needed for some models.
5. **Compare a baseline:** check whether the model beats a simple “next value equals last value” forecast.
6. **Avoid overclaiming:** stock prices are affected by news, economic events and investor behaviour.

8. Viva questions and model answers

Theory questions

1. What is time-series analysis? Analysing observations recorded sequentially over time to understand patterns and forecast future values.
2. Time-series vs cross-sectional data? Time-series follows variables over time; cross-sectional compares entities at one time or period.
3. What is stationarity? Broadly, statistical properties such as mean and variance remain stable over time.
4. What is autocorrelation? Relationship between a series and its own past values at different lags.
5. What is a trend? Long-term upward or downward direction.
6. What is seasonality? A pattern that repeats at a fixed interval.
7. Additive vs multiplicative decomposition? Sum of components versus product of components.
8. What is residual? The remaining variation after trend and seasonal components are removed.
9. What is exponential smoothing? A forecasting approach that weights recent observations more heavily.
10. What are train and test data? Training fits the model; testing evaluates predictions on unseen observations.

Code and interpretation questions

11. Why convert Date to datetime? To enable date-aware operations.
12. Why set Date as index? Resampling and time-series operations work naturally with a datetime index.
13. Why are early moving-average values NaN? Not enough observations exist for a full window.
14. What does 'ME' mean? Month End frequency for resampling.
15. Does monthly mean give month-end close? No, it gives average of observations in that month.
16. What does period=20 mean? A 20-observation cycle is assumed for decomposition.
17. Interpret 0.023473 correlation. Very weak positive linear association between closing price and volume.
18. Why use scatterplot? To see patterns, clusters and outliers.
19. What does trend='add' mean? Additive trend component.
20. What does forecast(30) return? Thirty future predictions.

9. Tricky cross-questions + speaking script

Questions examiners may use to test understanding

21. Does period=20 prove a 20-day cycle? No. It is an input assumption; the code does not prove that seasonality exists.
22. Why is a moving average delayed? Averaging dilutes sudden movements, so the smoothed line responds more slowly.
23. What is look-ahead bias? Accidentally using future information during model fitting or selection.
24. Why not train on test data? Then the test data is no longer unseen and evaluation becomes misleading.
25. What is MAE? Average absolute difference between actual and predicted values.
26. What is RMSE? Square root of average squared prediction errors; it penalizes large errors more.
27. Price vs return? Price is the quoted value; return measures change over a period, often as a percentage.
28. Can stock prices be predicted perfectly? No. Markets include uncertainty, news and unpredictable events.
29. Does correlation prove causation? No, it only measures association.
30. Why test on last 30 observations? This notebook chooses a 30-observation horizon; another horizon could be selected for a different use case.

60-second viva explanation

"This assignment performs time-series analysis on Apple's historical stock data. I imported pandas, NumPy, Matplotlib, Seaborn and the required statsmodels functions. I loaded the CSV using read_csv and inspected it using head. I converted Date to datetime, sorted chronologically and set Date as the index. I plotted AAPL.Close to inspect the historical trend. Then I calculated 20- and 50-observation moving averages using rolling().mean() to smooth short-term fluctuations. I resampled closing prices by month to calculate monthly averages and used seasonal_decompose with an additive model and period 20 to separate trend, seasonal and residual components. I calculated price-volume correlation, which was approximately 0.0235, indicating almost no linear relationship in this dataset. Finally, I trained an Exponential Smoothing model with additive trend on all but the last 30 observations and forecast 30 steps. The notebook plots actual and forecast values, but does not calculate numerical accuracy metrics such as MAE or RMSE."

Last-minute checklist

- ■ Explain datetime conversion and Date index.
- ■ Explain rolling versus resample.
- ■ Know why MA20/MA50 begin with NaN.
- ■ Define trend, seasonality and residual.
- ■ Interpret correlation 0.023473 accurately.
- ■ Explain chronological train/test split.
- ■ Explain trend='add' and forecast(30).
- ■ Mention the missing forecast metrics and frequency warning.

Source note

Code and reported correlation are taken from the uploaded Assignment_13.ipynb. The assignment sheet describes Assignment 13 as Time Series Data Analysis of stock prices and mentions ARIMA or exponential smoothing; the submitted notebook implements exponential smoothing only. Additional theory and improvement suggestions are provided for viva preparation.